

CMP5367: Artificial Intelligence and Machine Learning

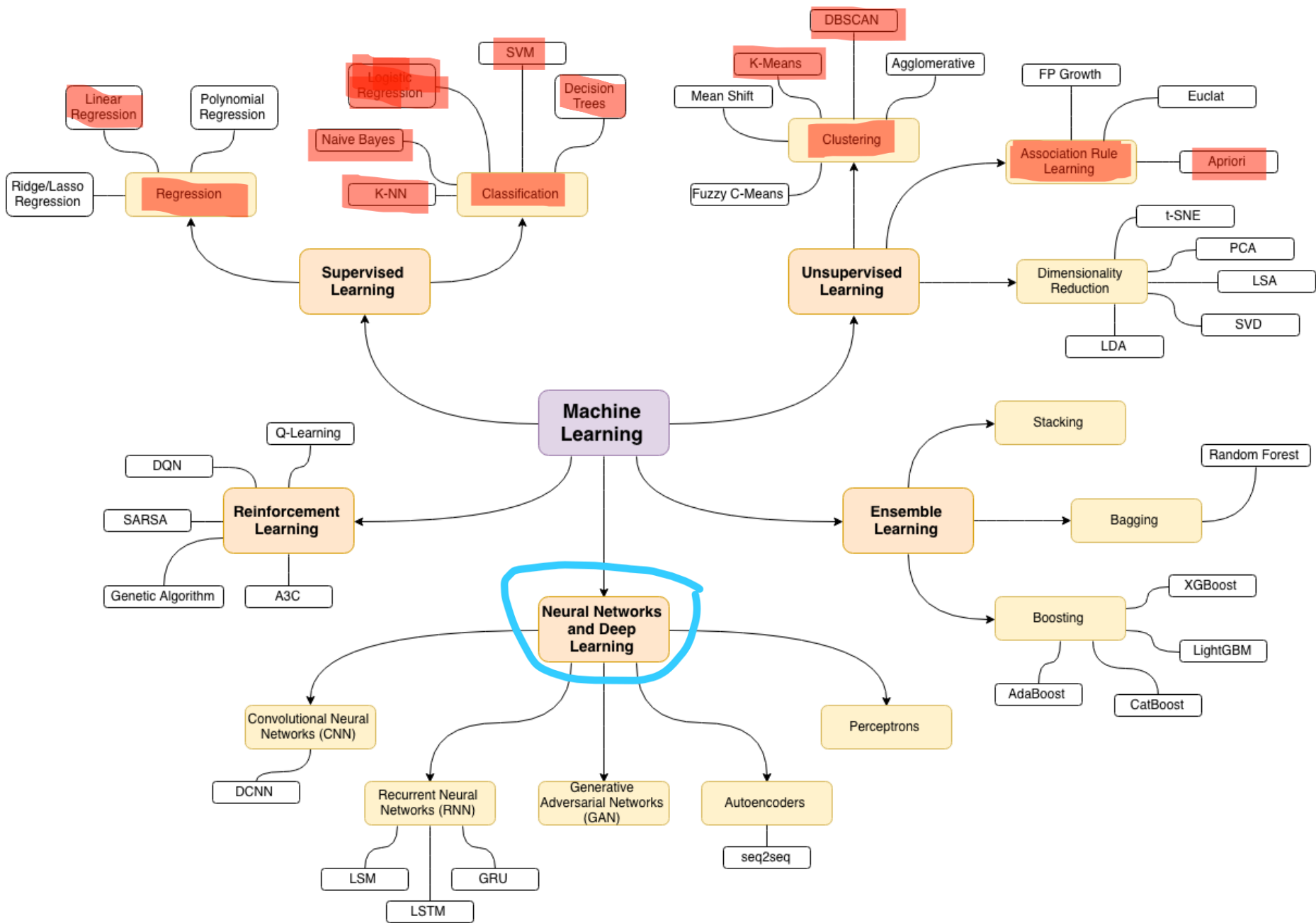
Lectures 7: Introduction to Deep Learning

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Outline

- Neural networks applications
- Deep learning intuition
- Forward and backward propagation
- Assessment 1 checking



Deep Learning: Neural Networks (NNs)

Machine Learning methods that has its roots dating back to the 1940's. Deep Learning gained attention in the last decades for its groundbreaking application in areas like image classification, speech recognition, and machine translation.



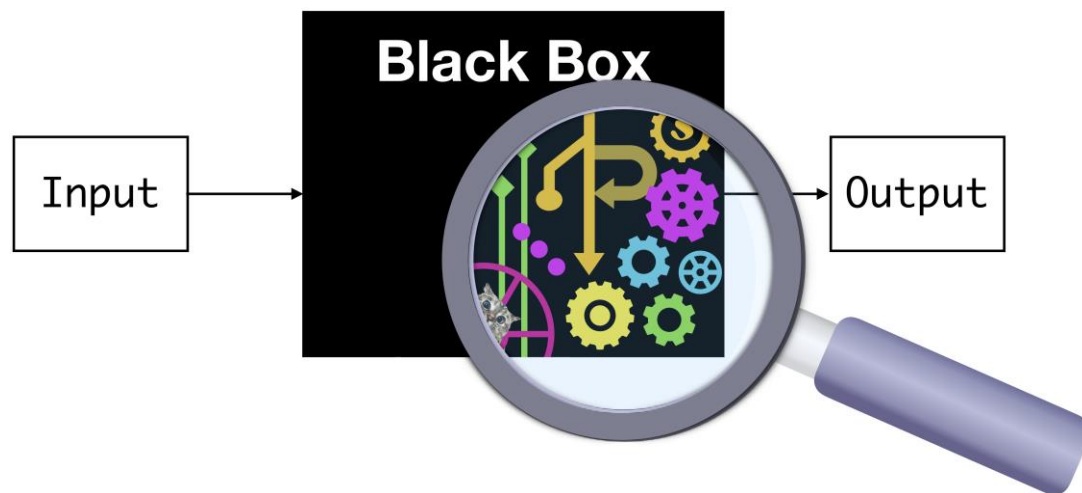
Applications of Deep Learning (DL)



Artificial Neural Networks (ANNs)

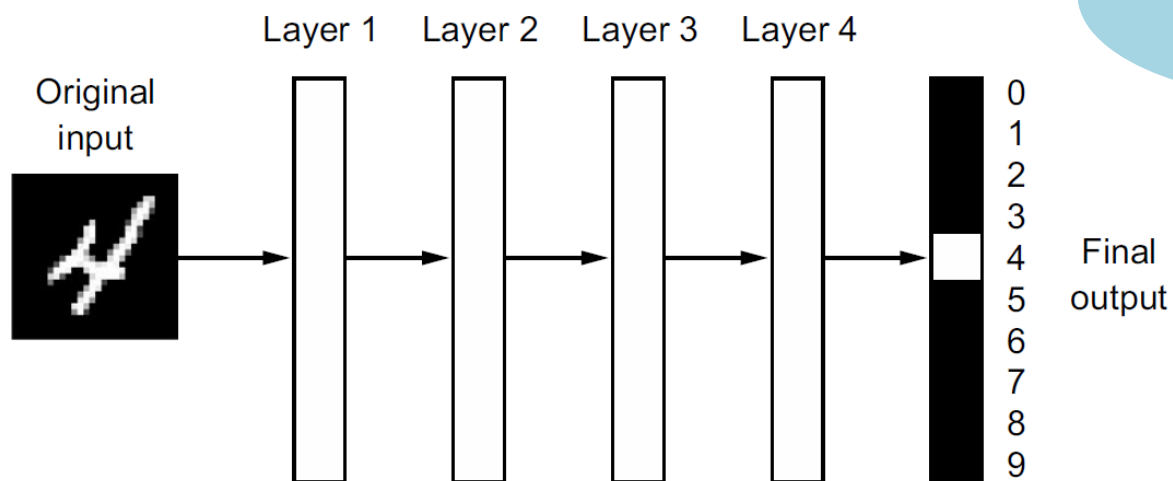
Deep Learning algorithms use [Artificial Neural Networks](#) :

- don't require expert input during the feature design and engineering phase.
- ANNs can *learn* the patterns of data.
- *learn* how to represent the data with features they extract on their own.
- without much human intervention in feature design and extraction, allows algorithms to adapt much faster to the data at hand.



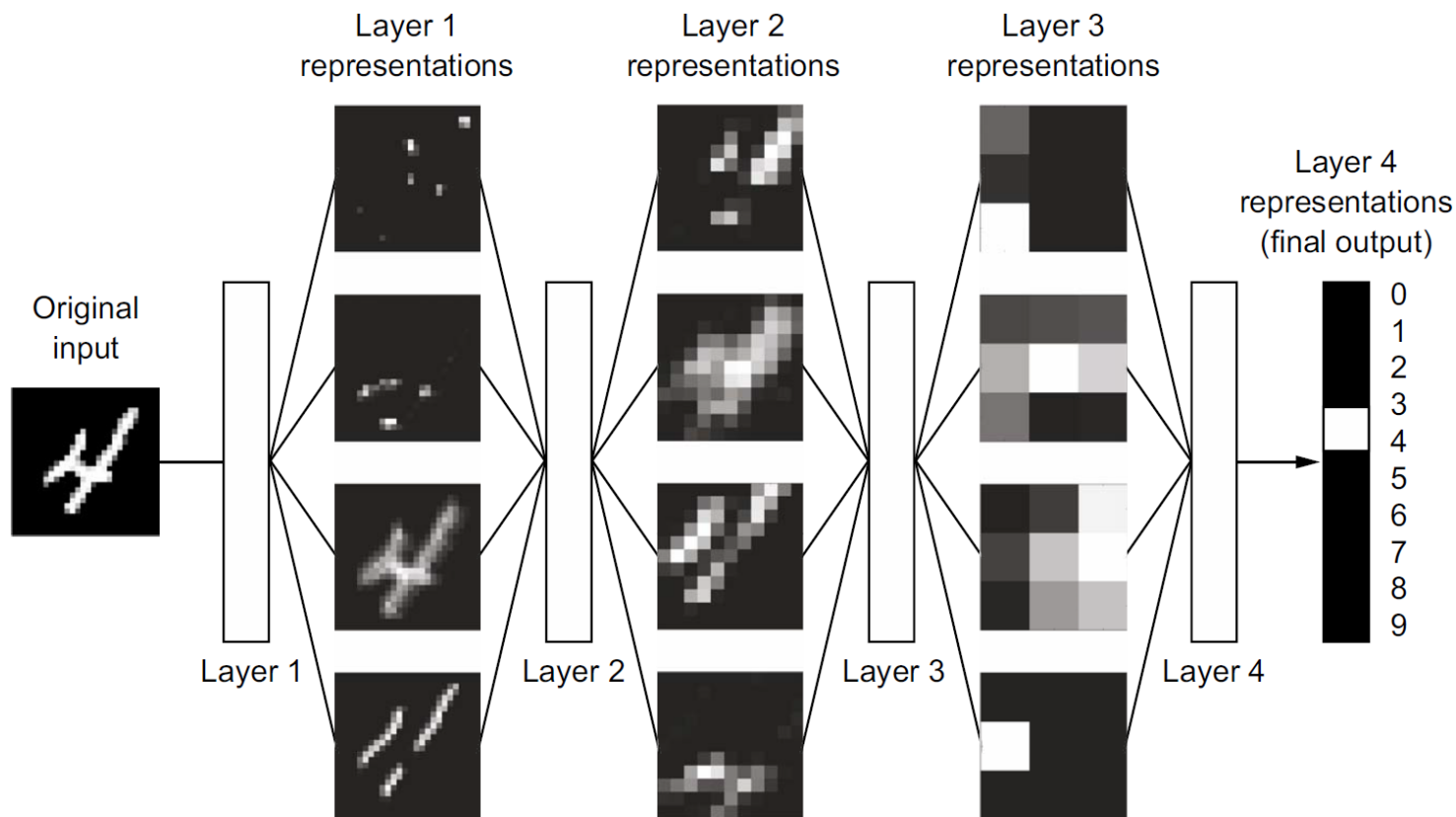
The “deep” in “deep learning”

The “deep” in “deep learning” isn’t a reference to any kind of deeper understanding achieved by the approach; rather, it stands for this idea of successive layers of representations.



A deep neural network for digit classification

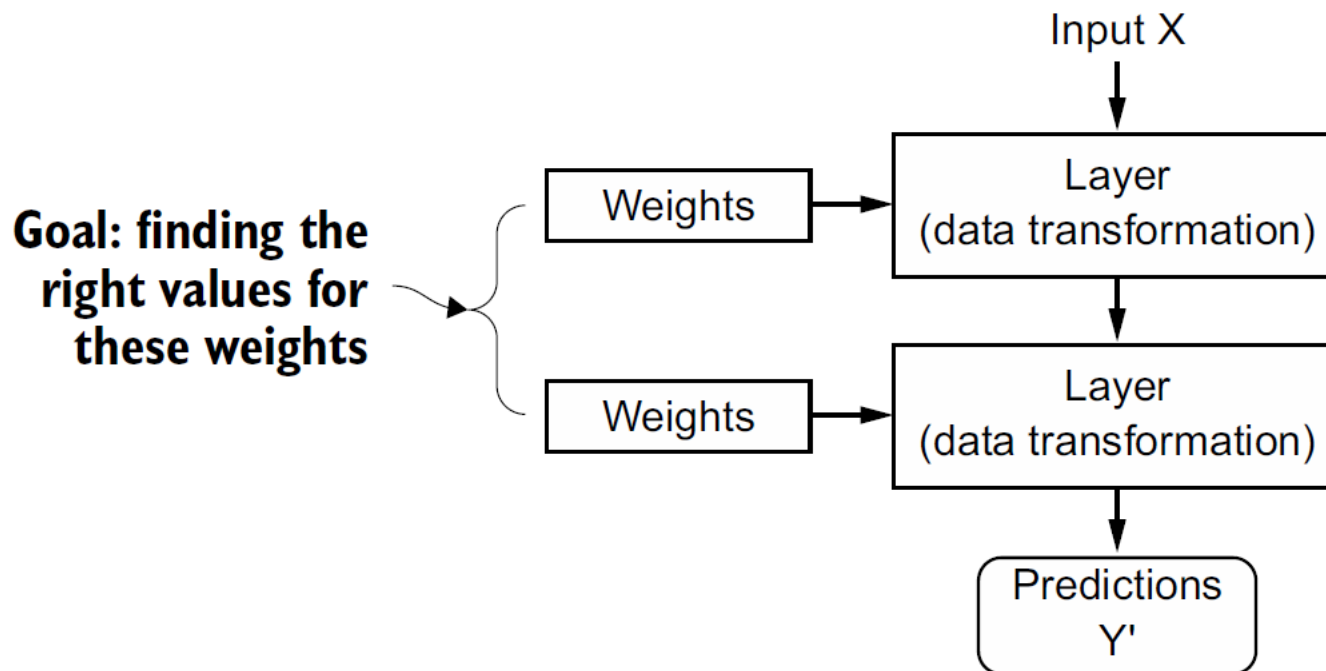
How the deep network works



So that's what deep learning is, technically: a multistage way to learn data representations.

Understanding how deep learning works, in three figures

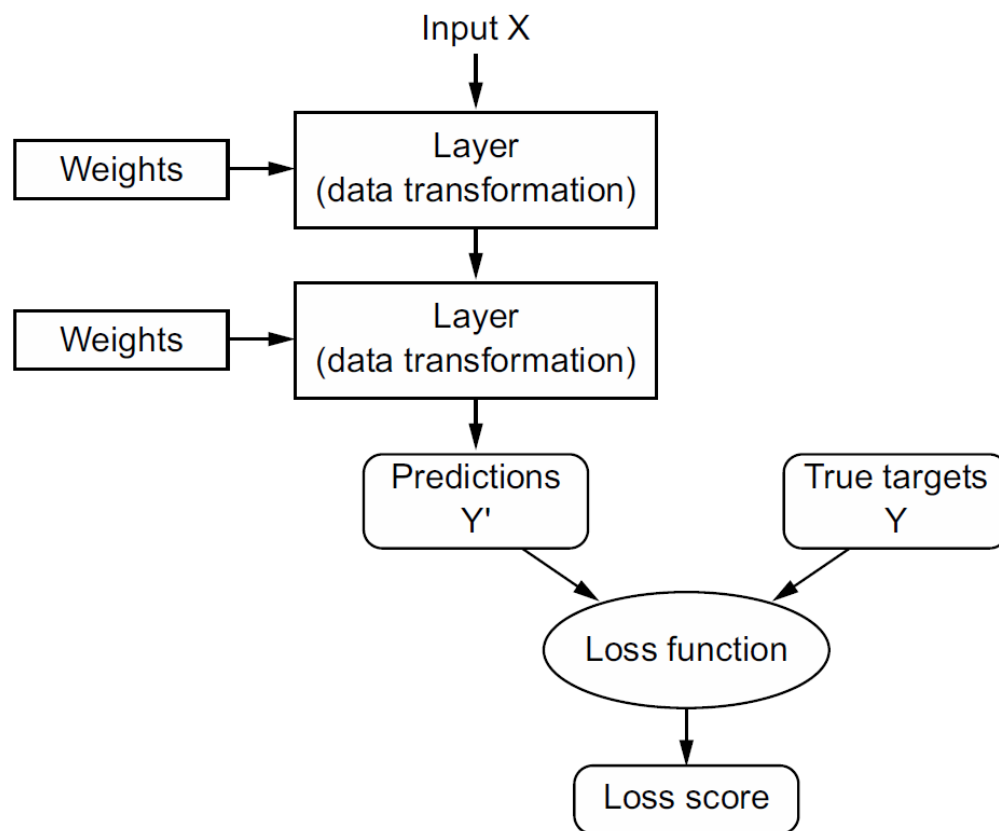
Figure 1



Learning means finding a set of values for the weights of all layers in a network, such that the network will correctly map example inputs to their associated targets.

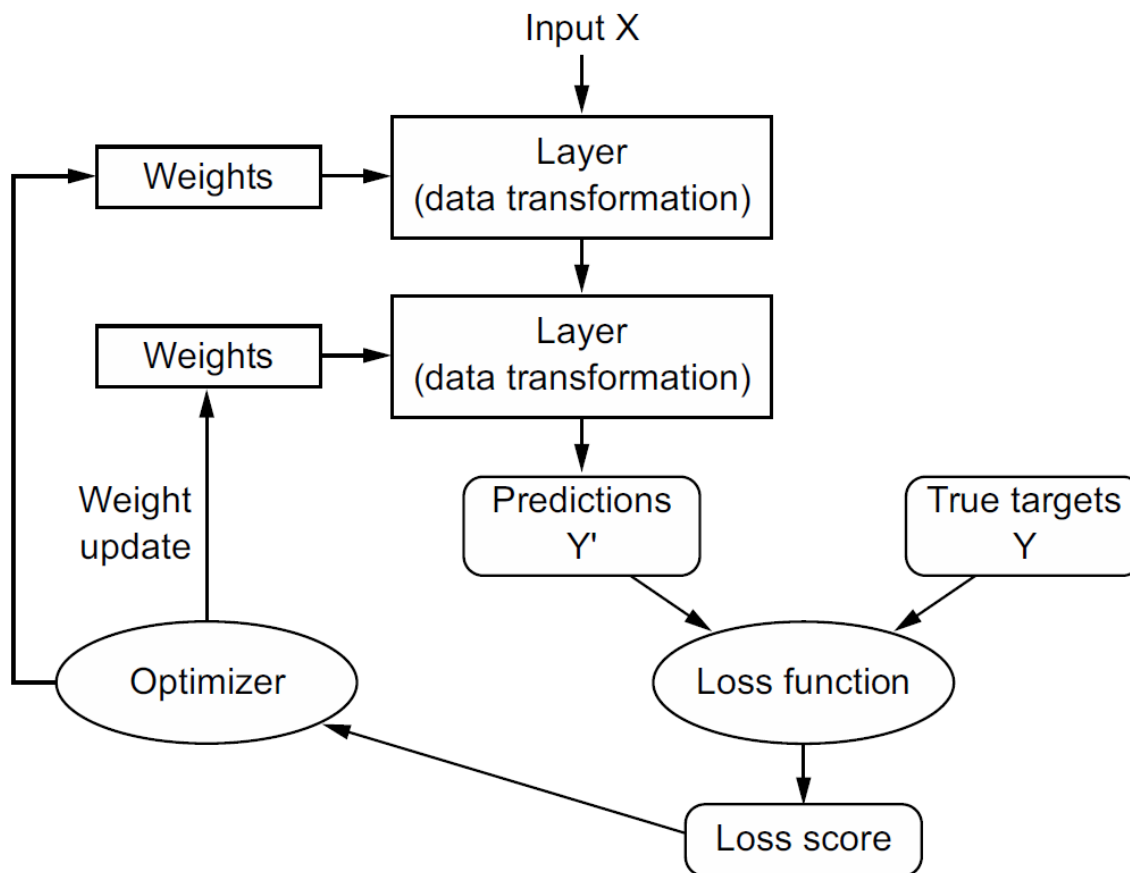
Weights are also called the **parameters** of a layer.

Figure 2



- A **loss function** measures the quality of the network's output.
- The **loss function** takes the predictions of the network and the true target (what you wanted the network to output) and computes a distance score.
- It measures how well the network has done on this specific example.

Figure 3

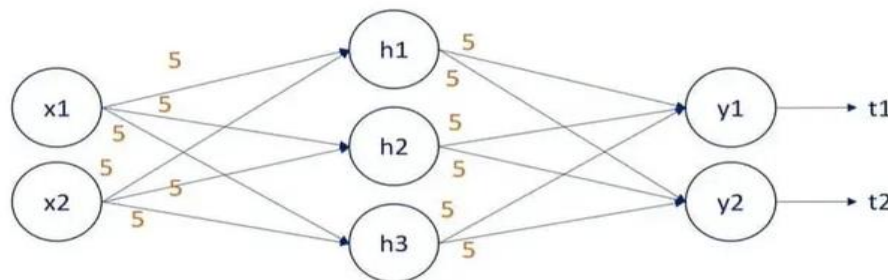


The fundamental trick in deep learning is to use *this score as a feedback signal to adjust the value of the weights a little, in a direction that will lower the loss score for the current example.*

The loss score is used as a feedback signal to adjust the weights.

How is the network initiated?

Initialization



- Initially, the weights of the network **are assigned random values**
- Naturally, its output is far from what it should ideally be, and the loss score is accordingly very high.
- But with every example **the weights are adjusted** a little in the correct direction, and the loss score decreases.
- This is what we call the **training loop**, which, repeated a sufficient number of times (typically tens of iterations over thousands of examples), yields weight values that minimize the loss function.
- A network with a minimal loss is one for which the outputs are as close as they can be to the targets.....This is what we call **a trained network**

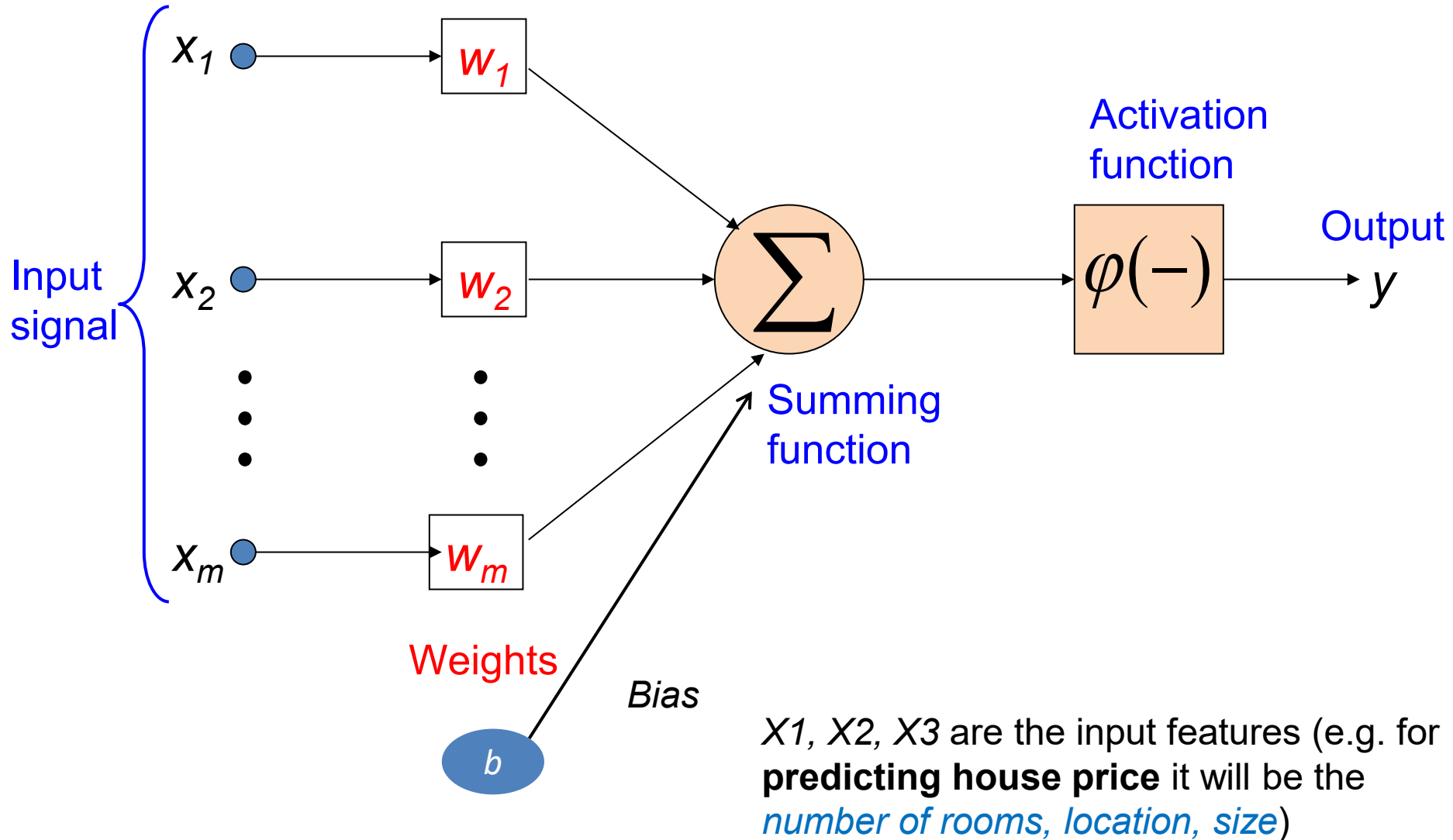
Quiz



How does deep learning work?

- Forward pass
- Backward pass

Stage 1: Forward Pass:

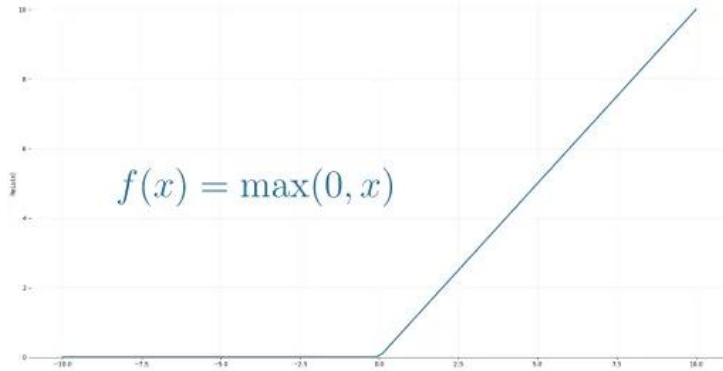


What is the activation function?

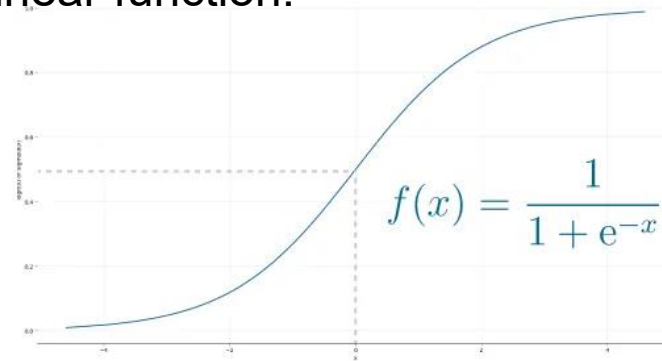
Activation functions introduce non-linearity into the network, allowing it to learn complex patterns and relationships in the data. **Otherwise, the model will only learn straight-line relationships**

Common activation functions include:

- **ReLU (Rectified Linear Unit):**
Returns 0 for negative inputs and the input itself for positive inputs.

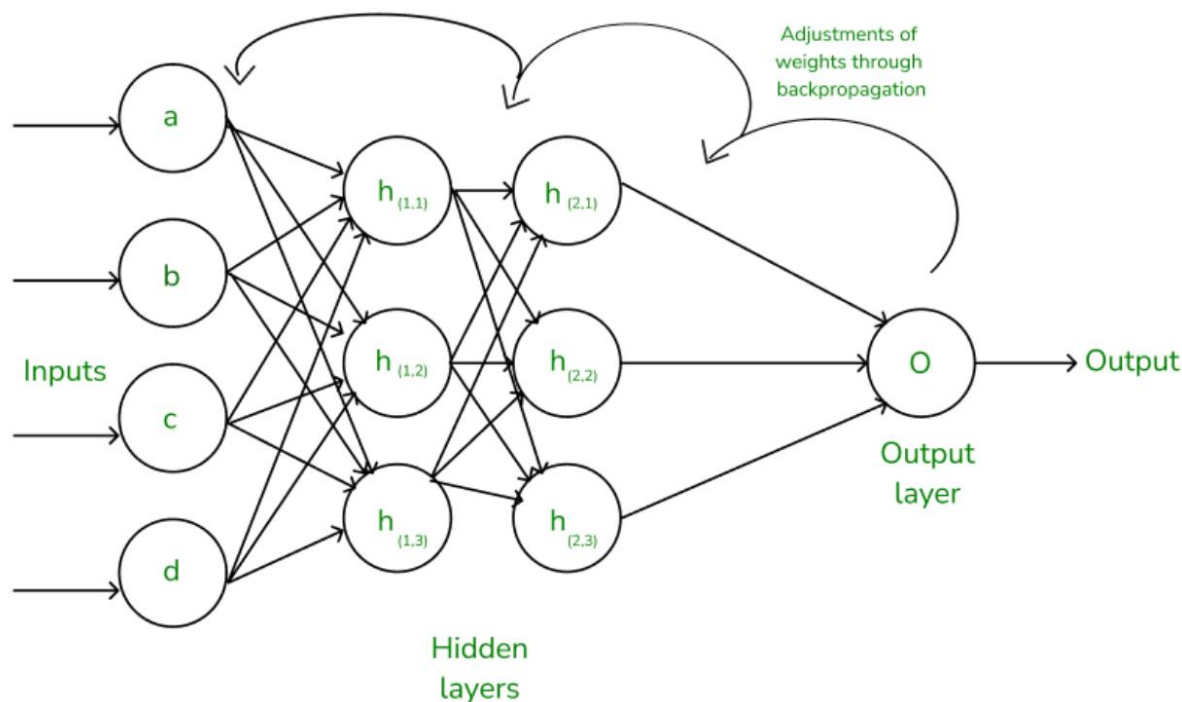


- **Sigmoid:** The sigmoid function maps any real input to a value that is either 0 or 1 and encodes a non-linear function.



The choice of activation function depends on factors such as the nature of the problem, the characteristics of the data, and empirical performance on validation sets.

Stage 2: backward pass (backpropagation)?



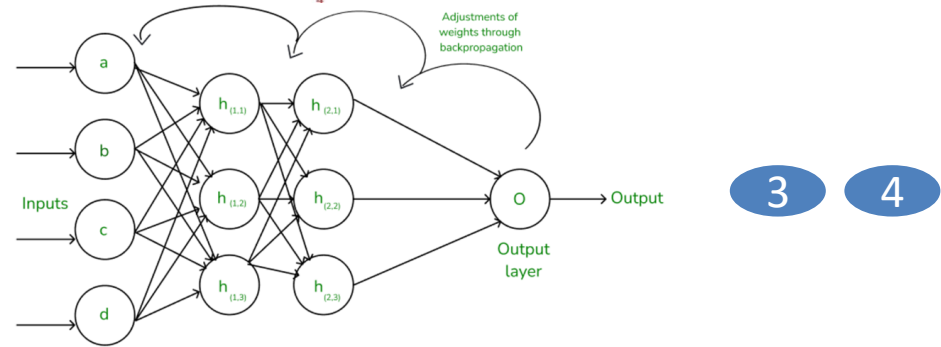
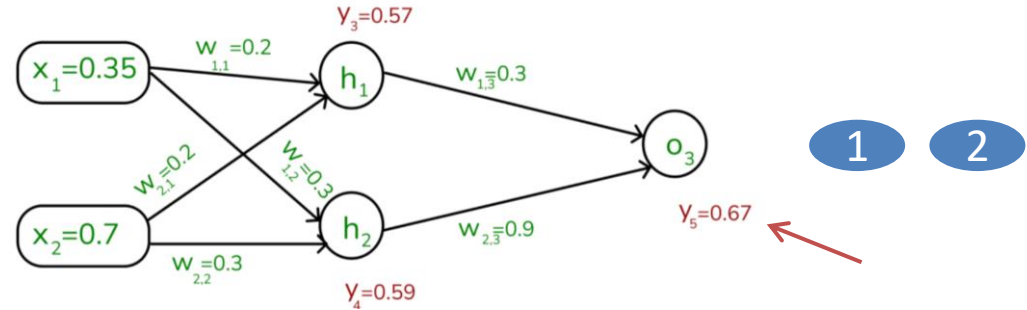
Backpropagation is the learning mechanism that allows the deep network to iteratively adjust the weights in the network, with the goal of minimizing the cost function.

Example

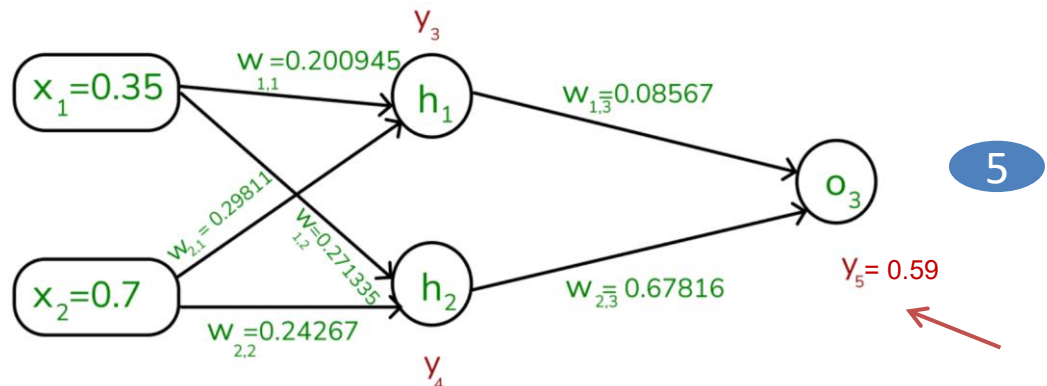
If my predicted values should be 0.5

1. Forward pass starts with random weights. Sum the feature values x with the weights w .
2. Then perform activation function (e.g. Relu) to get values of y (the hidden layer) then pass it to output function (O).
3. Then calculate the loss with a loss function and change the weights through an algorithm (gradient descent).
4. The weights are adjusted.
5. Keep repeating steps 1 to 4 till no improvement to score.

Forward pass1



Changed weights in back propagation



Backpropagation intuition

- During every **epoch (one full pass through the entire dataset)**:
 - The model sees **every training example once**.
 - It performs a **forward pass** to make predictions.
 - It calculates the **loss** (how wrong the predictions are).
 - It performs **backpropagation** to update the weights.

One epoch usually isn't enough for the model to learn well.

- Backpropagation optimization algorithms used such as gradient descent or stochastic gradient descent.
- *This gradient is responsible for telling us how much each weight/bias should be adjusted to minimize the error in the next forward pass.*

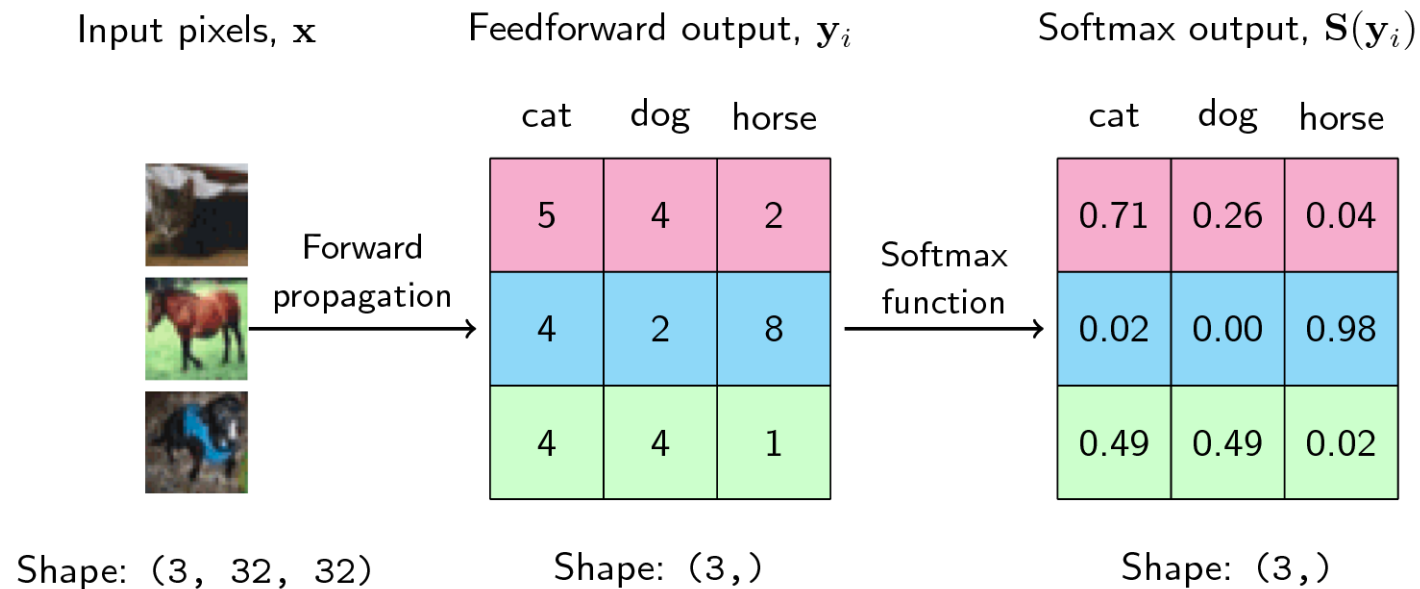
Forward and backward propagation animation!!

[Neural Network Visualizer | Interactive Deep Learning Tool](#)

[Backprop Explainer](#)

The output function:

- The weighted outputs from the last hidden layer are fed into the **last output** to compute the final prediction.
- The last layer uses an activation function (e.g. **softmax function**) which is responsible for converting the weighted outputs into probabilities for each class.



Output Functions

Task Type	Activation Function	Output Format	Example Use Case
Binary Classification	Sigmoid	Single scalar between 0 and 1	Spam detection, yes/no tasks
Multi-Class Classification	Softmax	Vector of probabilities (summing to 1)	Digit recognition (0–9)
Regression	None or Linear	Single scalar (any real number)	Predicting house prices
Multi-Label Classification	Sigmoid (per class)	Vector of independent probabilities	Tagging multiple topics in text

Main Types of Neural Networks

- **Convolutional Neural Networks (CNN):**
 - Specialized for processing grid-like data, such as images. They use convolutional layers to automatically learn hierarchical patterns.
- **Generative Adversarial Networks (GANs):**
 - Consist of two neural networks: a generator and a discriminator. They are trained simultaneously in a competitive setting, where the generator learns to generate realistic data, and the discriminator learns to distinguish between real and fake data.
- **Long Short-Term Memory (LSTM) Networks:**
 - A type of RNN that addresses the vanishing gradient problem and is capable of learning long-term dependencies in sequential data.

Neural Networks

- **Advantages**

- Prediction accuracy is generally high
- Robust, works when training examples contain errors
- Output may be discrete, real-valued, or a vector of several discrete or real-valued attributes
- Fast evaluation of the learned target function

- **Criticism**

- Long training time
- Difficult to understand the learned function (weights)
- Not easy to incorporate domain knowledge

THANK
YOU